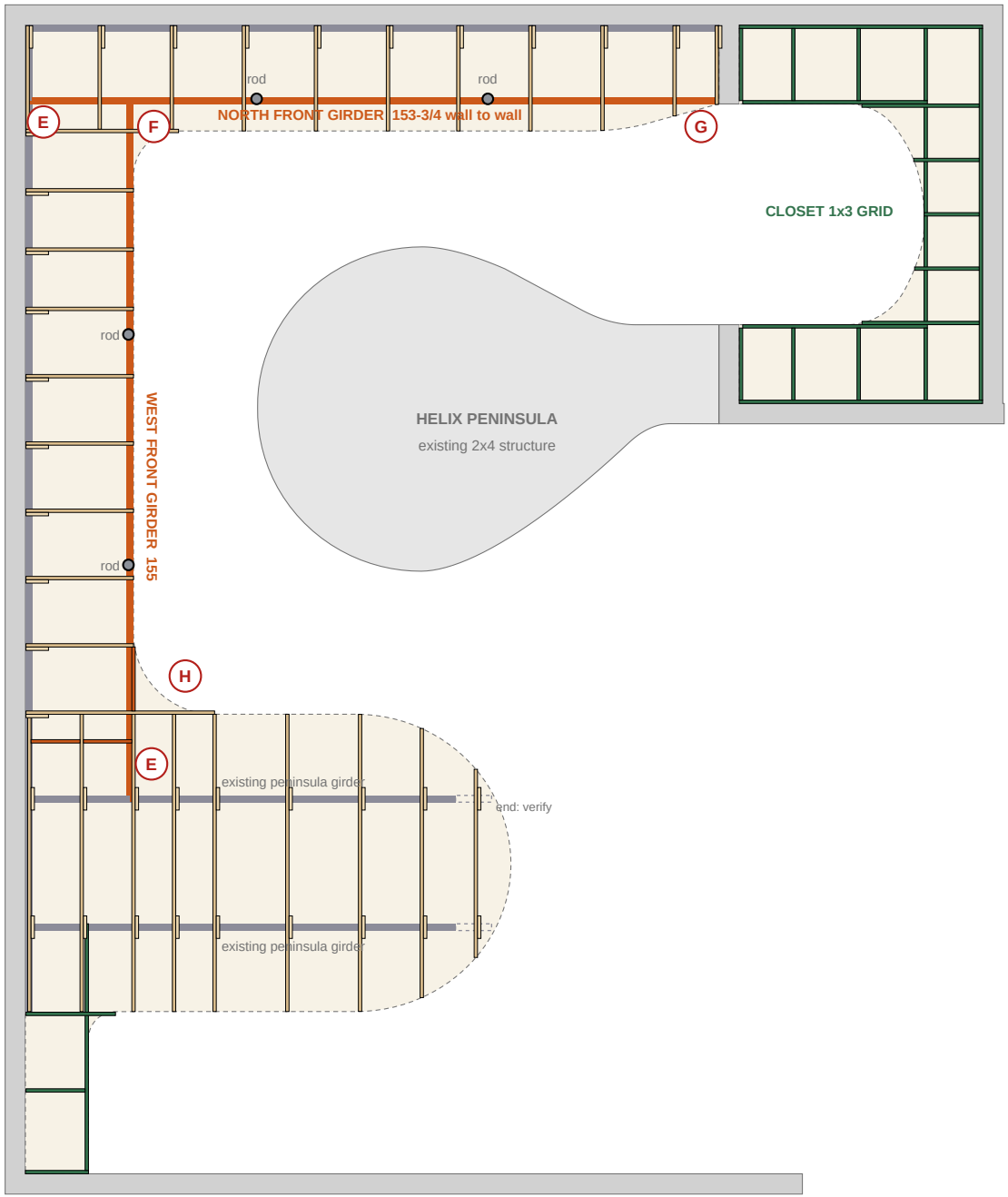
Framing plan - main level, deck removed

Looking down. Orange = new front girders; grey = existing girders.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



LEGEND

- New front girder (inverted L, 1x2 web on a 1x2 flange)
- New 1x2 joist
- 1x3 riser block on a -3.25 girder
- 1x2 blocking under a deck seam
- Closet 1x3 grid (ledger, rail, joist)
- Existing L-girder flange
- 5/16 rod, 2 per run, over a staging leg

NOTES

Joists run north-south on the north shelf, east-west on the west shelf and north-south on the SW peninsula, about 16 in o.c. Every deck seam lands on a joist or on blocking.

North girder set back to y 65 (face of the wall-stub end at the east end); the joists run on to the front edge, up to 6 in past it. West girder under the deck edge.

Rods: nominal at the third points (north x 56, 107; west y 116, 167). Move each one over a staging leg; keep every gap under 60 in.

E: front girder seats on an existing girder (sheet 4). F: west girder into the north girder. G: north girder at the wall stub. H: rim and blocking under F3.

Peninsula girders and joist ends at the tip: drawn to x 100; check the as-built girders before cutting the last two joists.

Closet grids are schematic: ledgers on the walls, a U of rails round the aisle, joists 12-16 in o.c.

JOIST STATIONS (centres, room inches)

North, x: 5, 21, 37, 53, 69, 85, 100-5/8, 116-5/8, 132-5/8, 148-5/8, 158

West, y: 71, 84-1/8, 97-3/8, 110-1/2, 125-1/2, 140-3/8, 155-1/4, 170-1/4, 185-1/4, 200-1/8

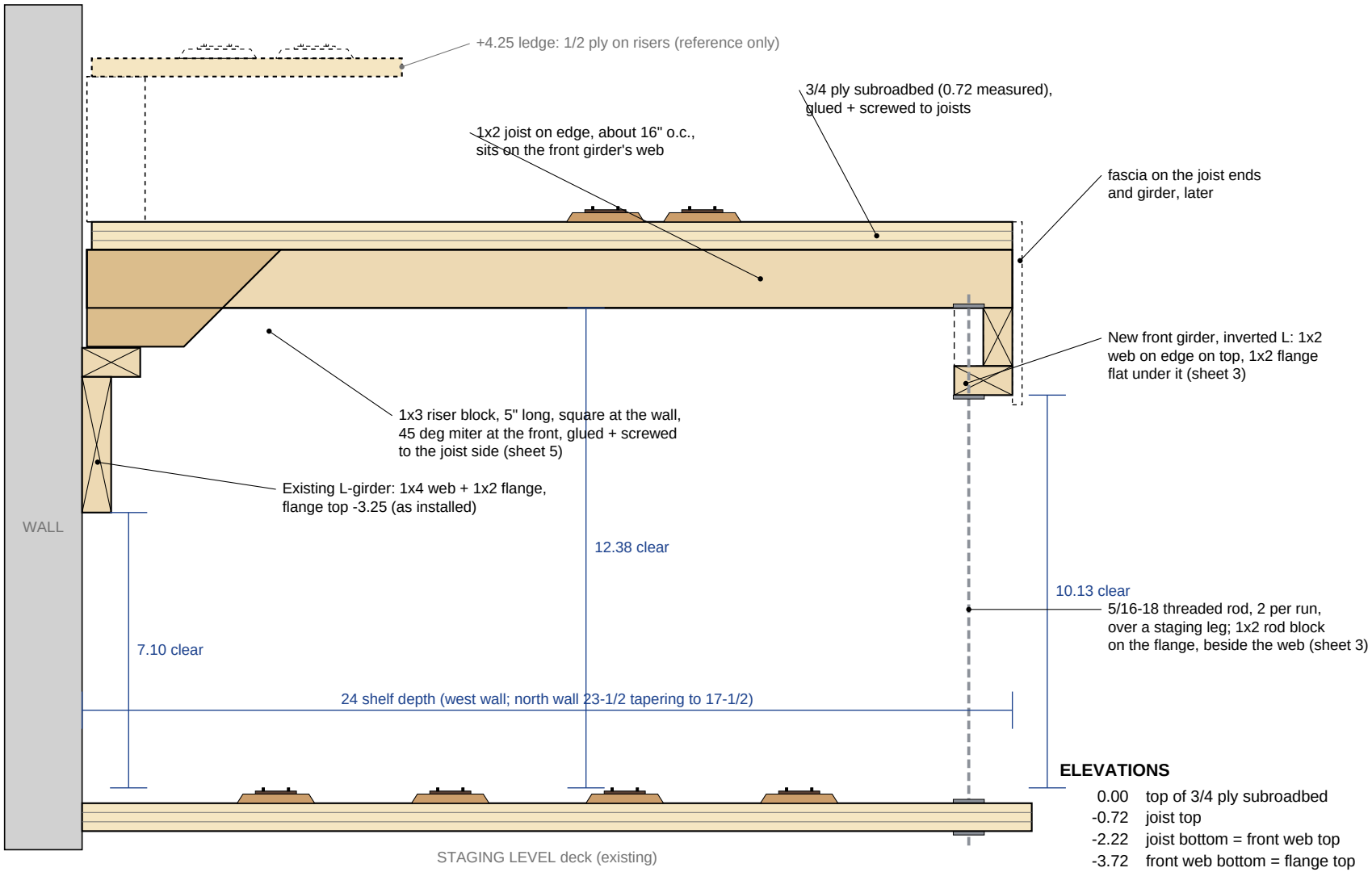
Peninsula, x: 5-3/8, 17, 28-1/2, 37-1/2, 46-1/2, 62-5/8, 78-3/4, 92-1/2, 104-1/2

Room inches: x east from the west wall corner, y south from the north wall line (same as the cut plan). Plan about 1:39 - work from the stations, not the drawing.

Typical shelf section - west wall (north wall the same, girder set back)

Section through a joist, perpendicular to the wall.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



ELEVATIONS

0.00	top of 3/4 ply subroadbed
-0.72	joist top
-2.22	joist bottom = front web top
-3.72	front web bottom = flange top
-4.47	front flange bottom
-3.22	riser block bottom
-3.25	back girder flange top
-7.50	back girder web bottom
-14.60	staging railhead
-15.00	staging deck top

Block bottom lands 0.03 above the back flange (3.25 - 2.50 - 0.72); the deck top finishes 1/32 low - within tolerance, or shim the flange. The joists sit straight on the front girder's web - no blocks at the front.

North wall: the same section with the girder set back to y 65, so the joists run past it to the front edge (up to 6 in). Track = 0.236 cork + 0.160 Atlas code 83.

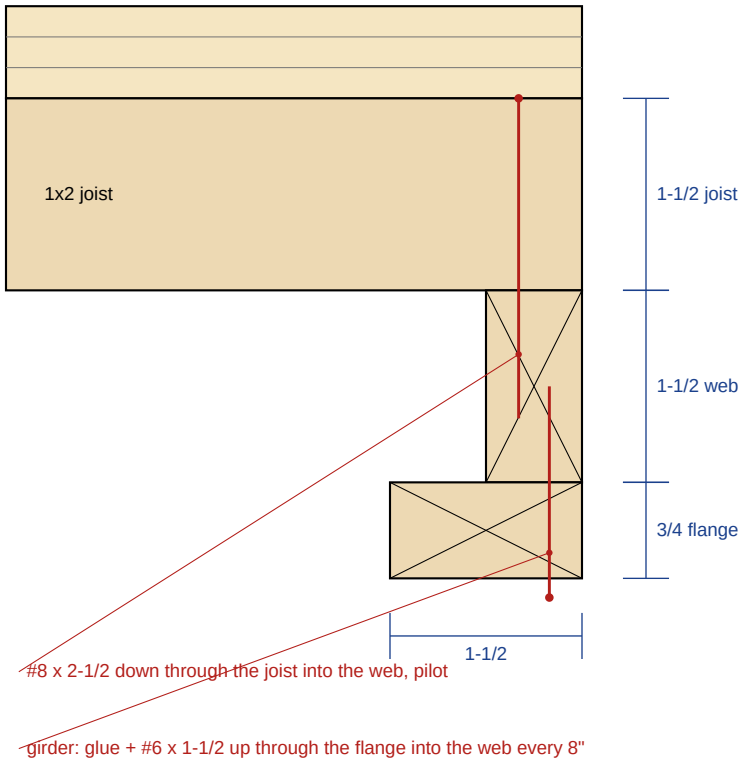
Scale 1:4 on Letter landscape. Lumber cut in section shows an X; the joist and riser block are in side view.

Details - front girder: inverted L, 1x2 web on a 1x2 flange

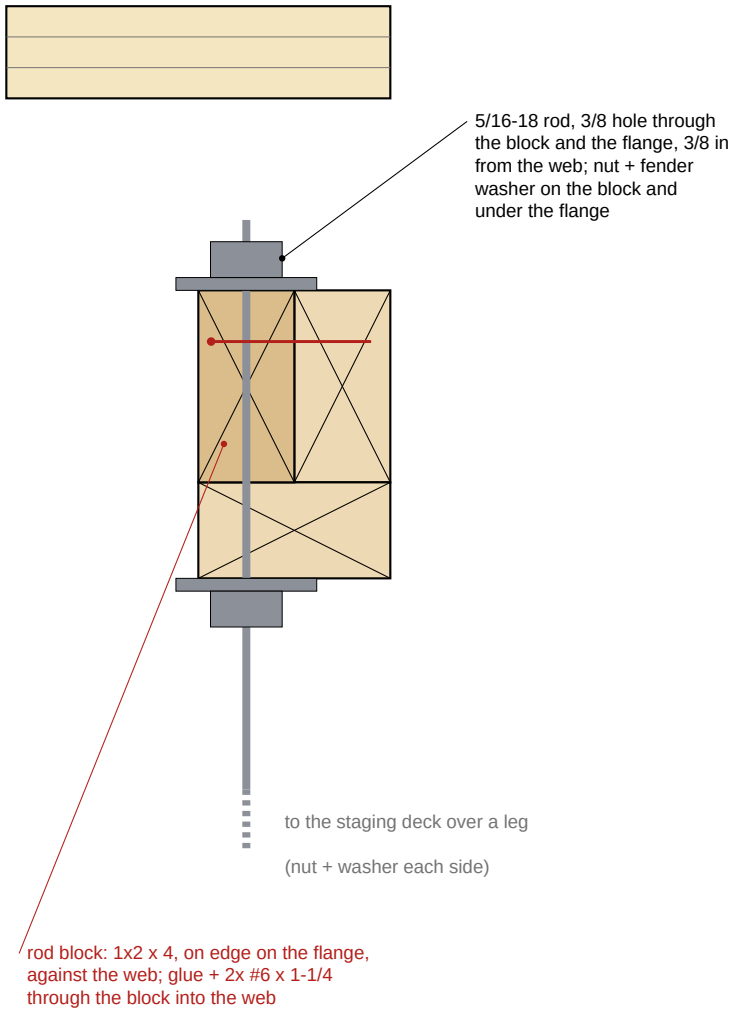
C: at a joist. D: between joists, at a rod. Both cut across the girder, looking along it.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

C AT A JOIST 1:1.5



D BETWEEN JOISTS, AT A ROD 1:1.5



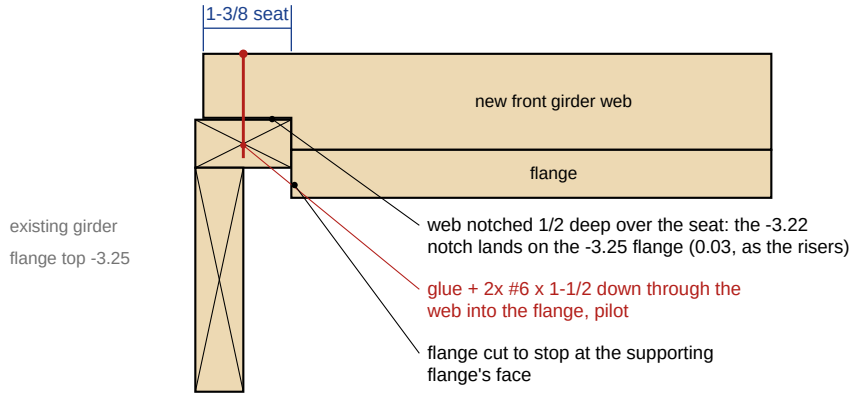
Build each girder on the bench: lay the flange flat, glue the web on edge on its outer half (outer faces flush), screw up through the flange every 8". North 153-3/4 and west 155 in: one 16 ft piece each, no splice. Before the joists go on, drill the rod holes where the staging legs fall, two per run, and fit the rod blocks. The web top is the joist bottom (-2.22); the joists brace the web's top edge.

Details - girder junctions

E: seat on an existing girder. F: west girder into the north girder. G: wall stub. H: under F3.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

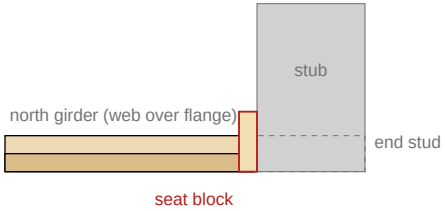
E SEAT ON AN EXISTING GIRDER 1:3



At the west wall: the north girder on the west wall girder, web end 1/8 off the wall.

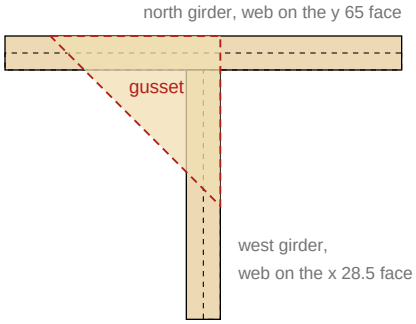
At the peninsula: the west girder across the peninsula's north girder flange (1-1/2 seat).

G NORTH GIRDER AT THE WALL STUB, PLAN 1:8



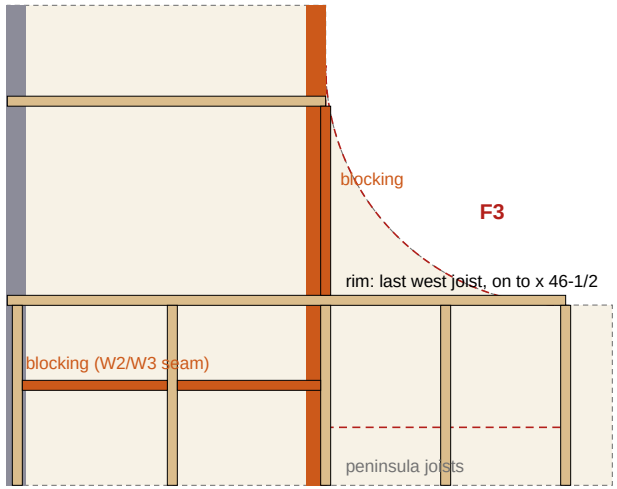
Seat block: 1x3 x 4 long, flat on the wall at the end stud, top at -4.47 under the flange end. 2x #8 x 2-1/2 through the block into the end stud; #8 x 3 down through the web and flange into the block. Girder end 1/8 off the wall. If there's no stud at the end of the stub, add blocking in the wall first.

F WEST GIRDER INTO THE NORTH GIRDER, FROM BELOW 1:8



1/2 ply gusset, 7-1/2 x 7-1/2 triangle, under both flanges: glue + 3x #6 x 1-1/4 up into each flange, pilot. Butt the west girder tight to the north girder's front face and glue it; 2x #6 x 2 through the north web into the end of the west web. Webs dashed: they sit above the flanges.

H RIM AND BLOCKING UNDER F3, PLAN 1:14



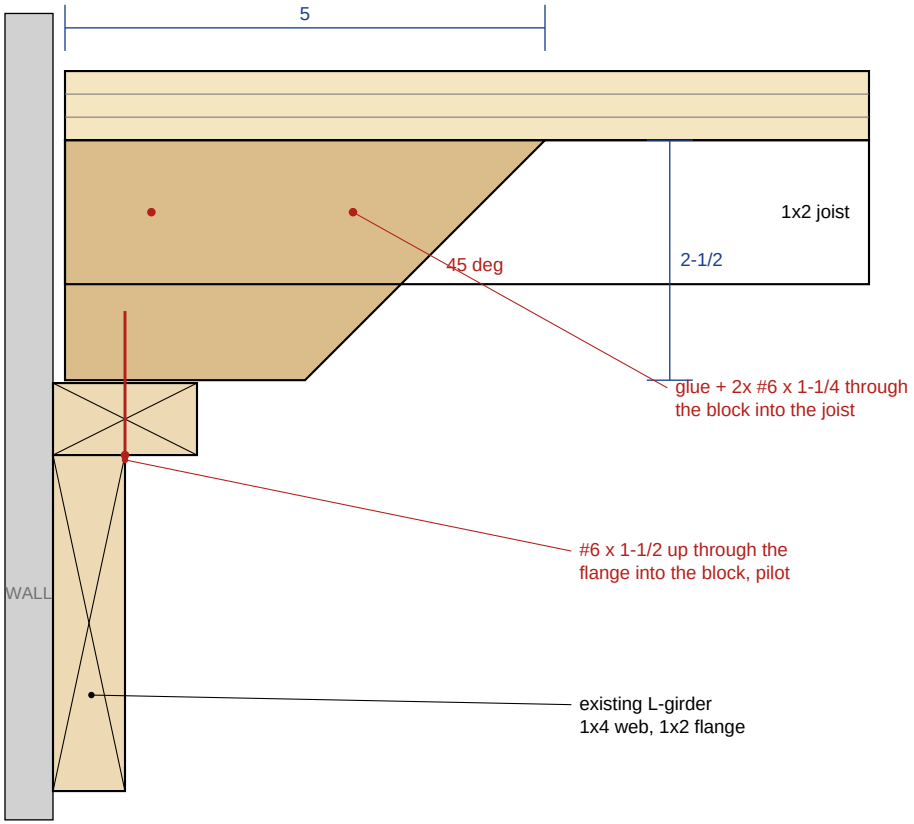
Peninsula joists butt the rim: 2x #6 x 2 through the rim into each end.
Blocking: 1x2 on edge between joists, ends screwed 2x #6 x 2.

Details - riser block at a -3.25 girder (back wall and peninsula)

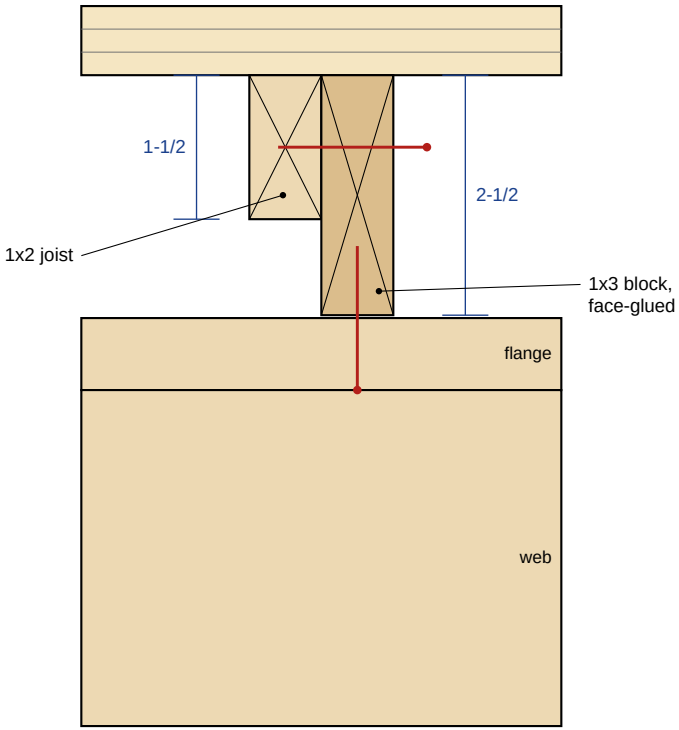
A: side view along the joist. B: looking along the joist (joist and block cut, girder in elevation).

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

A SIDE VIEW ALONG THE JOIST 1:2



B LOOKING ALONG THE JOIST 1:2

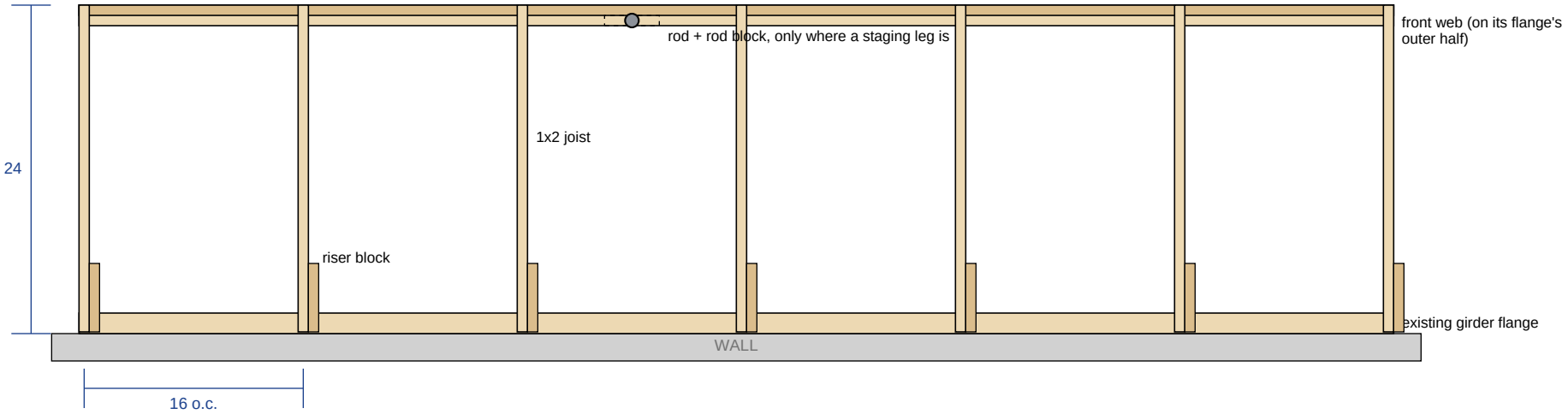


Riser block: cut from 1x3, grain running with the joist. Top flush with the joist top; the bottom sits on the flange. The wall end stays square; one full-height 45 deg miter on the front end leaves 2-1/2 of bottom, so the block still bears on the whole 1-1/2 flange. Joist and block ends sit 1/8 off the wall. On the peninsula girders the block is centred on the flange, miter toward the aisle. Same side on every joist. Pattern on sheet 6.

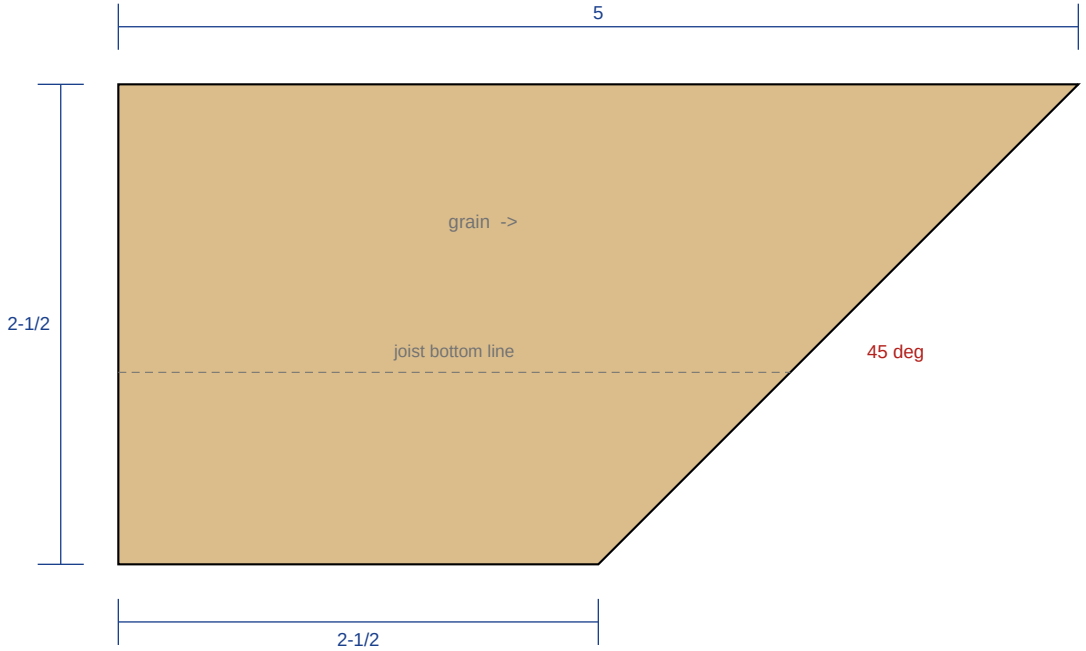
Plan - typical 8 ft bay, west wall, deck removed + riser block pattern

Looking down. Joists about 16" o.c.: riser blocks at the wall, straight onto the front web.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



RISER BLOCK - FULL SIZE PATTERN (1:1)



square end goes to the wall

PER 8 FT BAY (24" deep, west wall)

1x2 joists	7 @ 23-7/8	wall gap 1/8 to the deck edge
1x3 riser blocks	7 @ 5	square at wall, full 45 deg front
1x2 front web	96	on edge, joists on top
1x2 front flange	96	flat, under the web
3/4 ply	24 x 96	one rip per bay

The front girders go in as one 16 ft piece per run, no splice. Butt joints in the 3/4 ply land on a joist; the joist stations on sheet 1 already sit under every seam.

Scales: plan 1:12, pattern 1:1 - print at 100% and check the 5" dimension with a rule before tracing.

Assembly sequence, fasteners and lumber

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

1. Build both front girders on the bench as an inverted L: 1x2 flange flat at the bottom, 1x2 web on edge on its outer half. Glue + screw up every 8". One 16 ft piece each, no splice. Where a girder seats on another girder, notch the web end 1/2 deep and stop the flange at that girder's face (detail E).

2. Mark the staging legs under each front edge. Pick two per run, keeping every gap under 60". Drill 3/8 holes there and fit the rod blocks (detail D).

3. North girder: seat the west end on the west wall girder (E), the east end on the seat block at the wall-stub end stud (G). Prop at mid-span with the web top at -2.22.

4. West girder: seat the south end across the peninsula's north girder (E), butt the north end into the north girder and fit the gusset (F). Prop with the web top at -2.22.

5. Run the rods to the staging deck. Level both girders with the nuts, then lock with the opposing nuts.

6. Cut the riser blocks (pattern, sheet 6) and glue + screw one to each joist that lands on a -3.25 girder: the back wall end of every north and west joist, and both peninsula girders.

7. Set the joists at the sheet 1 stations; screw down through each joist into the front web, and up through the back flange into each block. The rim and the NW joist run on past the west girder (H).

8. Fit the seam blocking (H). Build the closet grids: ledgers into studs, the rail U, then the joists.

9. Dry-lay the 3/4 ply; seams on joists or blocking. Glue + screw every 8" along each joist, keeping screws out from under the track centrelines (feeders). Fascia last.

FASTENERS

Girder flange to web	glue + #6 x 1-1/2 @ 8"
Joist down into front web	#8 x 2-1/2, pilot
Block to joist	glue + 2x #6 x 1-1/4
Back flange up into block	#6 x 1-1/2, pilot
Rod block to web	glue + 2x #6 x 1-1/4
Seat (E), notched web	glue + 2x #6 x 1-1/2
Gusset (F)	glue + 3x #6 x 1-1/4 per flange
Seat block (G)	2x #8 x 2-1/2, 1x #8 x 3
Rim, blocking ends	2x #6 x 2
Ply to joists	glue + #6 x 1-1/4 @ 8"
Rods	5/16 nut + fender washer each side, 3/8 hole

LUMBER (from the framing plan)

1x2, 16 ft	4	front girder flanges + webs
1x2, 8 ft	13	30 joists, rim, blocking, rod blocks
1x3, 8 ft	9	39 riser blocks, closet grids
1x2 / 1x3 net	145 / 70 ft	before offcuts
5/16-18 rod	1 x 72"	4 pieces ~15"
Nuts / fender washers	16 / 16	
1/2 ply scrap	7-1/2 sq	gusset (F)

CLEARANCE OVER STAGING

Under joists	12.38 in
Under front girder	10.13 in
Under back girder web	7.10 in

Railhead = staging deck + 0.236 cork + 0.160 track.